

Transcript of Coursework

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July 9, 2026

Abstract

This document presents an overview of my studies in mathematics and computer science coursework conducted at Dartmouth College, with the addition of any self-guided and independent studies at the bottom. Only major related courses will be present (mathematics and computer science). **Grades marked with an * indicate currently enrolled courses.**

Keywords: probability, compressed sensing, machine learning, optimization.

1 Mathematics

Course 1.1 (MATH 71, Fall 2026). *Algebra*. Taught by Prof. Ina Petkova. [*]

Honors track introduction course to abstract algebra. Groups, Group actions, Lagrange's Theorem, Isomorphism Theorems, Sylow Theorems, Rings, Ideals, Quotient rings, Principal Ideal Domains, Unique Factorization Domains.

Course 1.2 (MATH 60, Spring 2026). *Honors Probability Theory*. Taught by Instructor Linh Huynh. [A]

Honors track first year probability theory course. Random variables, order statistics, convolutions, properties of expectation, moment generating functions, limit theorems, concentration inequalities, continuous-time Markov chains, stochastic optimization.

Textbook: Ross, *A First Year Course in Probability*; lecture notes.

Course 1.3 (MATH 24, Winter 2026). *Linear Algebra*. Taught by Prof. Dana Williams. [A-]

Dartmouth's equivalent of an abstract linear algebra class; proof variant of MATH 22 Linear Algebra with Applications. Vector spaces, linear operations, matrices, elementary matrix operations, determinants, spectral theory, inner product spaces, SVD.

Textbook: Friedberg, *Linear Algebra*; lecture notes.

Course 1.4 (MATH 11, Fall 2025). *Accelerated Multivariable Calculus*. Taught by Instructor Gayee Park. [A]

A fast paced introductory course covering MATH 9 and MATH 13 in one quarter. Vector geometry, space curves, limits and continuity, partial derivatives, tangent planes and differentials, multivariable chain rule, optimization problems, multiple integrals, Green's Theorem, Stoke's Theorem, Divergence Theorem.

Textbook: None.

2 Computer Science

Course 2.1 (COSC 61, Fall 2026). *Database Systems*. Taught by Instructor J. Peter Brady. [*]

This course introduces the principles behind how modern databases store, organize, query, and reliably manage data, with an emphasis on relational databases and transaction processing. SQL, Relational algebra, Database design, Entity-Relationship (ER) modeling, Normalization, Joins, Transactions, ACID properties, Indexing, Query optimization, Concurrency control, NoSQL/MongoDB.

Course 2.2 (COSC 50, Winter 2026). *Software Design & Implementation*. Taught by Instructor J. Peter Brady. [A]

An introductory systems programming course injected with lessons on software implementation. Covers Linux, bash commands, memory management, pointers, programming in C. Projects in C include Tiny Search Engine (a miniature web crawler), manually coding fundamental data structures, and an interactive server-client game.

Textbook: None

Course 2.3 (COSC 39, Fall 2025). *Theory of Computation*. Taught by Prof. Hsieh-Chih Chang. [*]

This course is an introduction to formal models of languages and computation. Topics covered include finite automata and regular languages, context-free languages, Turing machines and computability, NP-completeness and glimpses of computational complexity theory.

Course 2.4 (COSC 30, Winter 2026). *Discrete Mathematics for Computer Science*. Taught by Prof. Hsieh-Chih Chang. [A]

A discrete math class covering proof techniques (contradiction, induction, pigeonhole), counting, linearity of expectation, stars and bars, strong induction, graph proofs. Very competition math flavored course, with many interesting and fun problems!

Textbook: None

Course 2.5 (COSC 10, Fall 2025). *Problem Solving with Object Oriented Programming*. Taught by Prof. Alberto Quatrini Li. [A]

Now renamed to Data Structures and Reasoning. Course covers all major data structures/data types: arrays, linked lists, stacks, queues, binary trees, balanced trees, priority queues, hash tables, graphs; and major graph traversal algorithms: depth-first search, breadth-first search, Dijkstra's algorithm, A* search. In addition, miscellaneous algorithms/topics like Booyer Moore string matching, server client, concurrency and threads, and more.

Textbook: Goodrich, *Data Structures and Algorithms*.

3 Crosslisted Computer Science

Course 3.1 (ENGS 109, Spring 2026). *High-Dimensional Sensing and Learning*. Taught by Prof. Sang Chin. [A]

A graduate level compressed sensing course. Begins with sparse recovery theory: orthogonal matching pursuit, CoSaMP, subspace pursuit, restricted isometry property, mutual coherence, DCT and random sensing matrices, ℓ_1 minimization and convex relaxation. Ended with sparse matrix recovery, RPCA, proximal operators, non-negative matrix factorization, dictionary learning algorithms, IHT, LISTA. Research styled problem sets, and a poster presentation in addition to a written final.

Textbook: Foucart and Rauhut, *A Mathematical Introduction to Compressive Sensing*.

Remark. Implemented every matching pursuit algorithm from scratch before reading the reference implementation. Recommended.

Course 3.2 (ENGS 101, Fall 2026). *Principles of Reinforcement Learning*. Taught by Prof. Sang Chin. [*]

A graduate level course on reinforcement learning. Covers all major mathematical foundations, algorithms, and applications for reinforcement learning.